

# ETHAN CHANG

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## RESEARCH INTERESTS

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Machine Learning, Astrocyte-Informed Neural Networks, Reinforcement Learning, Computational Neuroscience

## EDUCATION

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**Washington University in St. Louis** - *St. Louis, MO* 2025 - Present  
*Ph.D. in Neuroscience*

**Johns Hopkins University** - *Baltimore, MD* 2025 - Present  
*M.S. in Computer Science*

**University of Rochester** - *Rochester, NY* 2021 - 2025  
*B.A. in Mathematics and B.S. in Neuroscience, High Distinction. Minor in Psychology*

## RESEARCH EXPERIENCE

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**Washington University in St. Louis** - *St. Louis, MO* 2025 - Present  
*Doctoral Researcher, Brain Dynamics and Control Research Group and Papouin Lab*

- Co-advised by ShiNung Ching, Ph.D. (Department of Electrical and Systems Engineering) and Thomas Papouin, Ph.D. (Department of Neuroscience)
- Investigating computational models that incorporate astrocyte-inspired reinforcement learning architectures for understanding neural computation and adaptive behavior.

**University of Rochester** - *Rochester, NY* 2021 - 2025  
*Undergraduate Researcher, Nedergaard Lab (Maiken Nedergaard, M.D., D.M.Sc.)*

- Conducted research on the role of aquaporin-4 localized on astrocytic endfeet in glymphatic transport dynamics.

## PUBLICATIONS

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- Gahn-Martinez D, Giannetto M, Chang E, Beam N, Pla V, Nedergaard M. Chronic intraventricular cannulation for the study of glymphatic transport. *eNeuro*. 2025. doi: 10.1523/ENEURO.0537-24.2025
- Giannetto M, Gomolka R, Gahn-Martinez D, Newbold E, Bork P, Chang E, Gresser M, Thompson T, Mori Y, Nedergaard M. Glymphatic fluid transport is suppressed by the AQP4 inhibitor AER-271. *Glia*. 2024. doi: 10.1002/glia.24515

## PRESENTATIONS

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- Chang E (presenter), Giannetto M, Agarwald I, Vento N, Gahn-Martinez D, Nedergaard M. Aquaporin-4 expression on glymphatic clearance routes and function. Schwartz Discover Scholar Showcase. 2024.
- Chang E (presenter), Giannetto M, Gahn-Martinez D, Nedergaard M. Aquaporin-4 expression and size-dependent solute movement in the brain. University of Rochester Undergraduate Research Exposition. 2024.
- Chang E (presenter), Barth RK. Isolation of hydrogen sulfide-producing bacteria from the environment. Department of Microbiology and Immunology Poster Session. 2023.
- Giannetto M, Gomolka R, Gahn-Martinez D, Newbold E, Bork P, Chang E (presenter), Gresser M, Thompson T, Mori Y, Nedergaard M. Glymphatic fluid transport is suppressed by the AQP4 inhibitor AER-271. University of Rochester Undergraduate Program in Biology and Medicine Poster Symposium. 2023.

## AWARDS AND FUNDING

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NIH T32 Predoctoral Trainee, Neuroscience Training Program (T32NS121881) 2025 - 2027

Schwartz Discover Grant 2024

## TECHNICAL SKILLS

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**Selected Programming/ML:** Python, Java, PyTorch, NumPy, pandas, scikit-learn, matplotlib

## TEACHING

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**Washington University in St. Louis, Department of Biology** 2026

*Assistant in Instruction, BIOL 3411: Principles of the Nervous System*

- Instructor: Keith Hengen, Ph.D and Mary Lambo, Ph.D.

**University of Rochester, Department of Brain and Cognitive Sciences**

2023

*Teaching Assistant, NSCI 201P: Basic Neurobiology Lab*

- Instructor: Renee Miller, Ph.D

## **SERVICE AND LEADERSHIP**

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**WashU Center for Career Engagement**

2025 - Present

*Graduate Student Advisory Board Representative*

- Provided graduate student input on career resources, especially for students interested in industry pathways.

**WashU St. Louis Neuroscience Outreach**

2025 - Present

*Student Volunteer*

- Participated in public neuroscience outreach at the Amazing Brain Carnival, St. Louis Science Center SciFest.

**University of Rochester Student Association**

2021 - 2025

*Deputy Chair for Academic Affairs Committee*

- Helped lead a team working on initiatives to improve the student academic experience and expand undergraduate research opportunities.

**University of Rochester Office of Undergraduate Research**

2023 - 2025

*Student Research Ambassador*

- Served as a student contact for research involvement and participated in admissions panels.

**Clinical and Emergency Service**

2021 - 2024

*Strong Memorial Hospital; RC-MERT*

- Emergency Department Clinical Support and Emergency Medical Technician roles in high-acuity care settings, including during the COVID-19 Omicron surge.

## **SELECTED COURSEWORK**

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**AI/ML:** Machine Learning; Advanced Machine Learning; Deep Learning; Probabilistic Graphical Models; Reasoning Under Uncertainty; Natural Language Processing

**Math/Statistics:** Probability; Mathematical Statistics; Stochastic Processes; Linear Algebra; Numerical Analysis; Multidimensional Calculus; Point-Set Topology

**CS/Systems:** Algorithms; Data Structures; Software Engineering; Computer Architecture; Computer Organization

## **LANGUAGES**

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English (native), Mandarin Chinese (conversational)